

HEMANTH RAJ PULUSU

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years of experience in product analytics, machine learning, experimentation, data engineering, and GenAI/RAG evaluation. Built product telemetry and log-analysis pipelines processing 3M+ daily events and 2M+ monthly records; improved model validation F1 by 9%, reduced manual review turnaround from 2-3 days to under 8 hours, and accelerated incident analysis by 35%. Strong hands-on experience with Python, SQL, Spark/PySpark, Scikit-learn, XGBoost, PyTorch, LangChain, LlamaIndex, vector search, AWS/GCP, Docker, MLflow, and BI tools. Known for translating ambiguous product questions into reliable metrics, model-ready features, A/B test plans, dashboards, and stakeholder-ready recommendations.

TECHNICAL SKILLS

- **Programming & Analytics:** Python, SQL, R, Pandas, NumPy, SciPy, Java, Bash
- **Statistics & Product Analytics:** A/B Testing, Hypothesis Testing, Confidence Intervals, Regression, Cohort Analysis, Funnel Analysis, Causal Inference, Time Series
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, Ranking Models, Feature Engineering, Model Evaluation
- **GenAI, NLP & LLM Evaluation:** LangChain, LlamaIndex, RAG, Agentic Workflows, OpenAI/Gemini APIs, ChromaDB, FAISS, RAGAS, Prompt Evaluation
- **Deep Learning:** PyTorch, TensorFlow, MONAI, 3D U-Net, CNN, LSTM, Hugging Face Transformers, CUDA
- **Data Engineering:** Spark, PySpark, Kafka, Hadoop, Hive, Airflow, ETL Pipelines, Data Quality, Data Modeling
- **Cloud & MLOps:** AWS (S3, EC2, Lambda, RDS, IAM, SageMaker), GCP (Cloud Run, BigQuery, Vertex AI, Cloud Storage), Docker, Kubernetes, GitHub Actions, Jenkins, MLflow
- **Visualization & BI:** Tableau, Power BI, Streamlit, Plotly/Dash, Matplotlib, Seaborn, Grafana

PROFESSIONAL EXPERIENCE

Data Scientist

Apple Inc, Austin, TX | September 2025 - Present

- Own Python and SQL feature pipelines across 5+ product telemetry sources, processing 3M+ daily interaction events into engagement KPI tables, model-ready features, and monitoring dashboards used in weekly product reviews.
- Built classification and ranking models with XGBoost and Scikit-learn; improved validation F1 by 9% over baseline models through feature diagnostics, error analysis, threshold tuning, and segment-level performance checks.
- Performed cohort, funnel, and regression analyses on engagement and review-quality metrics, helping product teams identify high-impact user segments and reduce follow-up analysis cycles by about 20%.
- Designed analysis plans for 3 A/B experiments, defining primary metrics, guardrail metrics, sample-size assumptions, 95% confidence intervals, and launch-readout summaries for stakeholder decision-making.
- Implemented an enterprise-approved LLM classification and summarization workflow for 1,000+ weekly review items, reducing manual review turnaround from 2-3 days to under 8 hours with human QA sampling and precision/recall tracking.
- Automated 7 weekly KPI dashboards and 12+ data-quality checks for schema, freshness, null-rate, and outlier monitoring, saving 6-8 analyst hours per week and reducing late-stage data investigations by roughly 20%.

Graduate Teaching Assistant

Florida Atlantic University, Boca Raton, FL | August 2023 - May 2025

- Led 24+ lab sessions for 60+ students across algorithms, data structures, OOP, and programming fundamentals, maintaining a 95% student satisfaction rating across two semesters.
- Created 10+ programming assignments, grading rubrics, and Git-based submission workflows; reduced grading turnaround from about 5 days to 48 hours while improving reproducibility and code review quality.
- Reviewed 350+ coding submissions and 40+ project deliverables, tracking recurring error patterns and using office-hour feedback to improve lab completion, debugging quality, and assignment readiness.

Software Engineer - Data & Platform Engineering

DXC Technology, Bangalore, India | June 2020 - June 2023

- Built SQL and Python pipelines to process 2M+ monthly production log records across enterprise applications, identifying anomalies and reducing mean incident resolution time by 35%.
- Developed log-based anomaly detection and pattern reports using thresholds, trend analysis, and service-level segmentation, reducing recurring alert noise by 18% and improving root-cause triage for support teams.
- Migrated data processing assets for 9 applications to AWS (S3, EC2, IAM), improving access control, availability, and operational reliability while reducing manual infrastructure support effort.
- Automated deployment and data validation pipelines with Jenkins and Bash, supporting 12 consecutive production releases with zero data-integrity defects reported after deployment.
- Implemented 20+ validation checks for schema drift, completeness, duplicates, load freshness, and record counts, reducing recurring data-quality investigations by roughly 25%.

PROJECTS

Agentic RAG Knowledge Assistant - LLM Evaluation Platform

Python, LangChain, LlamaIndex, OpenAI/Gemini APIs, FastAPI, ChromaDB/FAISS, RAGAS, MLflow, Docker, GCP Cloud Run, Streamlit

- Built an agentic RAG assistant over 200K+ documents with retrieval, planning, citation generation, and human-feedback loops, reducing query resolution time from 8 minutes to 45 seconds in test workflows.

- Designed router, retriever, evaluator, and summarizer agents using semantic chunking, embeddings, metadata filters, reranking, prompt templates, and guardrails, achieved 0.87 RAGAS faithfulness and 65% lower hallucination rate versus a naive LLM baseline.
- Tracked latency, answer quality, retrieval relevance, and user feedback across 100+ evaluation prompts in MLflow and Streamlit dashboards, creating a repeatable evaluation process for production-readiness decisions.

OwlTA ClassBuilder - 5-Agent Generative AI Orchestration System

OpenAI API, LangChain, Flask REST API, Discord.py, Prompt Pattern Catalog, Python

- Built a 5-agent LLM workflow that generated lesson plans, quizzes, and class activities in under 30 seconds, cutting sample content-preparation time by about 70% while keeping a teacher-controlled review loop.

3D Brain Tumor Segmentation - Custom U-Net Deep Learning Model

PyTorch, MONAI, NIfTI/nibabel, Dice Score, IoU, Sensitivity, Specificity, CUDA

- Trained a custom 3D U-Net encoder-decoder on 484 BraTS MRI scans and implemented Dice, IoU, Sensitivity, and Specificity tracking to evaluate tumor sub-region segmentation quality.

Cloud-Native AI Image Management Platform

Gemini Vision API, Google Cloud Run, Cloud Storage, Docker, Cloud Build, GitHub Actions

- Deployed a serverless image-intelligence app with auto-generated titles/descriptions, blue-green CI/CD, GCP IAM controls, and zero-downtime rollouts across 3 production versions.

EDUCATION

- Master of Science in Data Science and Analytics | Florida Atlantic University | May 2025
- Bachelor of Technology in Electronics and Communication Engineering | Jawaharlal Nehru Technological University Hyderabad (JNTUH), Hyderabad, India | 2016 - 2020

CERTIFICATIONS & COURSEWORK

- AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services
- Big Data Analytics Certification | Florida Atlantic University
- Google Professional Machine Learning Engineer (2026)